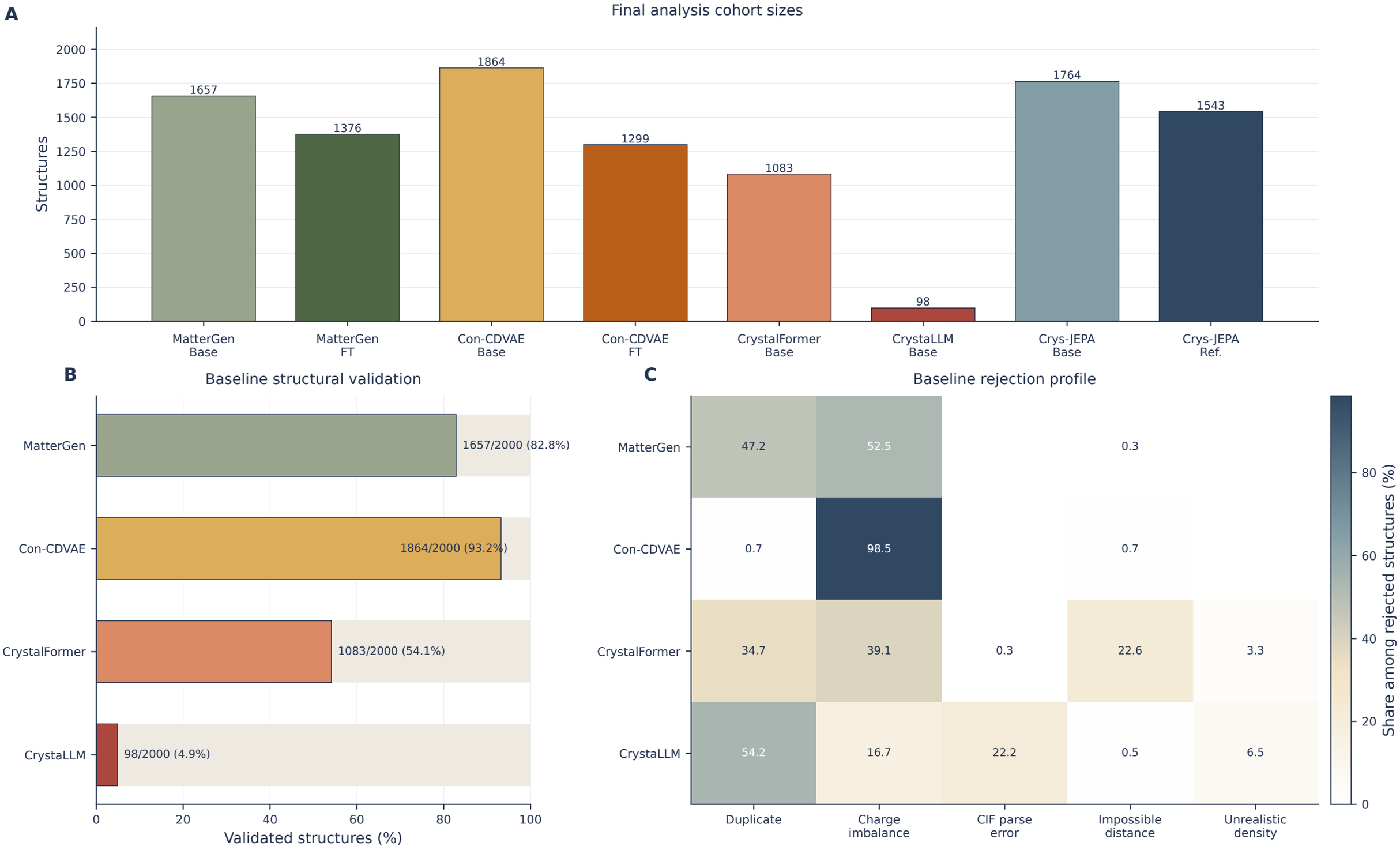
A
Final analysis cohort sizes
Structures
2000
1750
1500
1250
1000
750
500
250
0
1657
1376
1864
1299
1083
98
1764
1543
MatterGen Base
MatterGen FT
Con-CDVAE Base
Con-CDVAE FT
CrystalFormer Base
CrystaLLM Base
Crys-JEPA Base
Crys-JEPA Ref.
B
Baseline structural validation
MatterGen
Con-CDVAE
CrystalFormer
CrystaLLM
1657/2000 (82.8%)
1864/2000 (93.2%)
1083/2000 (54.1%)
98/2000 (4.9%)
0
20
40
60
80
100
Validated structures (%)
C
Baseline rejection profile
MatterGen
Con-CDVAE
CrystalFormer
CrystaLLM
47.2
52.5
0.3
0.7
98.5
0.7
34.7
39.1
0.3
22.6
3.3
54.2
16.7
22.2
0.5
6.5
Duplicate
Charge imbalance
CIF parse error
Impossible distance
Unrealistic density
Share among rejected structures (%)
0
20
40
60
80